



Panel Data Models for Predicting Penalty Kick Outcomes: A Discrete Choice Framework

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LIST OF ABBREVIATIONS

- IFAB: International Football Association Board
- FIFA: Fédération Internationale de Football Association
- DCM(s): Discrete Choice Model(s)
- MNL: Multinomial Logit (model)
- IIA: Independence of Irrelevant Alternatives
- ABAB, ABBA: Penalty shootout sequence formats
- ASC: Alternative-Specific Constants.
- AIC: Akaike Information Criterion
- BIC: Bayesian Information Criterion
- FE: Fixed Effects
- LR test: Likelihood Ratio test
- p-value: Probability value (statistical significance measure)
- GK: Goalkeeper
- ID: Identification Number (Player_ID)
- SE: Standard Error
- Rho^2 : McFadden's rho-squared fit statistic
- NL: Nested Logit (model)
- ML: Mixed Logit (model)

Abstract

This report investigates the decision-making processes behind penalty kick shot selection in elite football, focusing on the roles of player footedness, situational context, and temporal behavioral patterns. Using an unbalanced panel dataset of 2,358 professionally recorded penalties from international tournaments and top-league matches, six target zones of the goal were modeled as mutually exclusive alternatives within a discrete choice framework. The Multinomial Logit model served as the primary analytical tool, extended progressively to include inertia effects, player-specific fixed effects, contextual covariates, and interaction terms. Comparative exploration of Mixed Logit and Nested Logit models addressed heterogeneity and correlation between alternatives.

Empirical results revealed a consistent anti-inertia effect: elite players are significantly more likely to change target zones between consecutive kicks, highlighting the strategic advantage of unpredictability. Player identity was the dominant factor, with individual baseline preferences often overriding general trends; in contrast, footedness and goalkeeper reputation had negligible marginal effects once player fixed effects were included. Preferred locations were typically the upper and lower corners, especially under high-pressure shootout conditions, whereas match-deciding situations prompted avoidance of technically risky zones such as the top center. The final model achieved 42.24% predictive accuracy—five percentage points above a frequency-based naive benchmark—while maintaining interpretability. A dedicated panel data analysis by footedness compared penalty outcomes of left- and right-footed players, revealing cross-body shooting tendencies and confirming that right-footers and left-footers display distinct but consistent directional biases over time. An interactive dashboard was developed to visualize these aggregate and player-specific patterns, enabling side-by-side comparison of individual histories, footedness trends, and outcome distributions for use by coaches, analysts, and media.

This study contributes to sports analytics by demonstrating a structured, stepwise modeling strategy for behavioral prediction in high variance performance contexts, combining robustness with applied relevance. A Streamlit-based interactive application was developed to translate model outputs into an accessible decision-support tool. Limitations include the sparsity of individual player histories and collinearity between some tactical variables, constraining more complex forms such as Mixed Logit. Future research should leverage larger, more balanced datasets to refine heterogeneity modeling and explore richer interaction effects, potentially enhancing both predictive power and tactical usability.

1 Introduction

Football, the world's greatest game, is regulated by strict rules established by the International Football Association Board (IFAB). One of its most stressful situations is the penalty kick, which is awarded when a foul worth a direct free kick is made inside a team's penalty area. The situation creates a direct one-to-one showdown between shooter and keeper from 12 yards (11 meters), with the keeper the only defender.

Penalties are taken in two contexts: as part of regular play for box fouls, and as shootouts to break tied knockout games after extra time. In shootouts, teams take turns with five attempts each; if still tied, the match proceeds in sudden death. Shots are normally split by both direction (left, center, right) and elevation (high or low), producing six main target areas. Although they appear straightforward, penalties involve sophisticated strategic and mental decisions depending on the player's characteristics, goalkeeper mentality, situation in the match, and prior performance. FIFA statistics report that 75%–80% of penalties in top-level competitions are successful, but success is varied with these variables.

The study explores penalty kicking using Discrete Choice Models (DCMs), namely the Multinomial Logit (MNL) model, to account for the direction of a shot. Historical data from top competitions, the FIFA World Cup and UEFA Champions League, serve as the context in which to estimate the "utility" of each shot option. Player factors such as footedness, situation pressure, and movement of the goalkeeper are included in the analysis, using Biogeme, a free Python software for discrete choice modeling.

Footedness is one of the major variables to emerge. The stronger foot of a player dictates biomechanics strike mechanics, body position, approach angle, and positioning of the supporting leg of which impact direction, flight path, and power of the shot. Most players exhibit directional bias with their dominant foot, and goalkeepers attempt to predict these tendencies. Kickers can deliberately counteract these trends, however, to mislead the keeper, overlaying a psychological factor on the physical one.

To quantify these influences, we used a panel MNL model, controlling for contextual pressures and goalkeeper movement. While the MNL provided a good starting point, its assumption of Independence of Irrelevant Alternatives (IIA) and the sharing of the same parameter estimates across players limited its ability to capture natural variability and correlation among similar shot alternatives.

To get around this, we estimated a Mixed Logit model, which relaxes the IIA assumption and deals with unobserved heterogeneity by permitting important coefficients such as footedness and goalie movement to have random player specific values. This gave more sensitive behavioral insight but was technically awkward, with convergence problems, collinearity, and estimates instability, particularly with our unbalanced panel data.

As a less unstable alternative, we considered the Nested Logit model, which retains the interpretability of the MNL but groups similar shot options into "nests" (left/right, high/low). This structure allows for a twostep choice process selecting a general area of the goal before targeting a precise zone whilst permitting correlations among the nests. Lastly, we applied the Nested Logit with inertia effects for further analysis. Although Mixed Logit was theoretically important, its computational demands were higher than their advantages in this specific case. The exercise with Mixed Logit though shaped the building of the Nested Logit approach,

impacting variable selection and model specification so that it is balanced between analytical intensity and tractability.

2 Problem Description

The penalty kick is one of the most decisive moments in football, with the ability to settle closely matched games. Essentially, it is a contained, high stakes duel between an attacking player and the goalkeeper, taken from a stationary distance of 11 meters. By removing the dynamic fluidity of open play, the penalty provides a controlled setting in which both involved parties are forced to make quick, concurrent strategic decisions. For the kicker, the paramount decision is where to aim i.e., top left or bottom right corner while the goalkeeper needs to choose whether, and in which direction, to dive. These decisions are made under conditions of uncertainty and are influenced by psychological, physical, and situational variables.

In penalty shootouts, the stakes are even higher, deciding the winner in knockout matches that remain drawn after extra time. The two sides take five alternated shots, with sudden death splitting the tie if the scores remain level. Rules are simple: the ball must move forward, and the goalkeeper must remain on the goal line until contact time. Disregarding rare rule violations, penalties have one of three outcomes:

- **Goal:** The ball crosses the line into the net.
- **Save:** The goalkeeper stops the ball from going into the net.
- **Miss:** The ball does not go into the goal without a save.

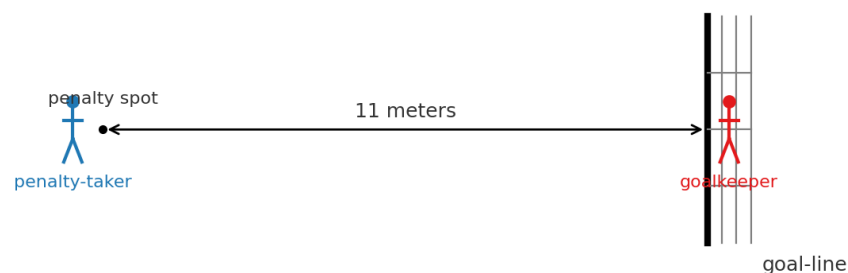


Figure 1: How a Penalty is shot?

With its well-defined alternatives and limited decision space, the penalty kick is a natural fit for Discrete Choice Modeling (DCM), specifically the Multinomial Logit (MNL) framework. The shooter's selection of six potential target areas can be represented probabilistically based on observable explanatory factors. Here, the data comprises player footedness, goalkeeper movement tendencies, as well as performance metrics. The main objective is to model and forecast shot direction while uncovering systematic behavioral patterns particularly those influenced by a player's dominant foot and determining how these patterns converge with goalkeeper behavior.

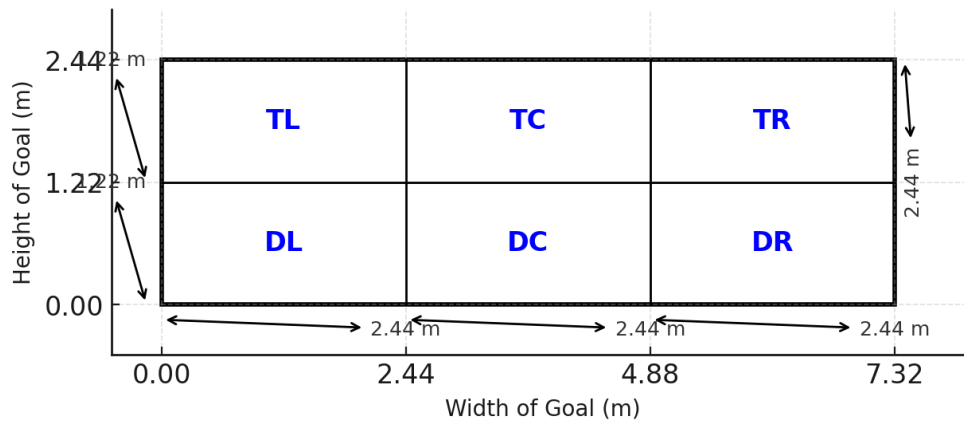


Figure 2: Soccer goal with the considered alternatives

The research is guided by the question: In what way does a player's preferred foot (right or left) influence penalty outcomes goals, saves, and misses at the elite level? The question addresses whether there is a systematic effect of footedness on shot placement and success rates, and whether these effects are consistent for repeated attempts within the same player. The analysis is guided by three interrelated sub questions:

1. Shot choice differentiation: Do right and left footed players differ significantly in their shot location probabilities, after controlling for repeated observations and individual effects?
2. Aggregate outcome alignment; Are penalty outcomes by footedness observed to be consistent with the patterns suggested by the panel MNL model, providing intuitive validation of its estimates?
3. Practical presentation; Is it possible for the findings to be presented in an interactive, easy to use way for both technical and nontechnical groups of coaches, scouts, and journalists?

A number of analytical issues present themselves. Professional football is dominated by right footed players, leading to imbalanced sample sizes across categories and possible inference problems. Penalty attempts are seldom independent; player choice is influenced by habit, learned tactics, and psychological toughness, each of which needs to be controlled for in the modeling exercise. Further, statistical output like coefficients, t values, and loglikelihoods, while informative to experts, can be hard to decipher for practitioners. Closing this gap involves a translation of technical findings into intelligible, actionable results.

To address these challenges, the study integrates three complementary pillars:

- **Robust modeling:** A panel MNL framework capable of estimating choice specific constants and isolating footedness effects in repeated measures data.
- **Descriptive context:** Aggregated summary statistics and graphs to give an intuitive overview of footedness category differences.
- **Interactive analytics:** A visual dashboard allowing investigation of both aggregate trends and player specific patterns without the need for sophisticated econometric expertise.

The MNL model is a natural choice: its discrete choice form is compatible with the mutually exclusive character of penalty shot locations, and its estimation is tractable and interpretable. This positions it well for an application in which the goal is not just to quantify the effects of determinants such as footedness and goalkeeper motion but also to convey the results clearly to audiences who are not statistical experts.

3 Data Collection, Preparation and Analysis

In this section, we present the collected dataset, detailing the process of identifying potential predictor variables, the sources used for data compilation, and the procedures applied to ensure high data quality. We also provide a brief analysis of the dataset, highlighting key statistics and summarizing the most relevant findings.

3.1 Literature Review

The literature on penalty kicks offers valuable insights into the psychological, strategic, and situational factors influencing penalty taker behavior. Penalty kicks have been used as a lens for examining decision-making under pressure, with approaches spanning game theory, behavioral economics, empirical modeling, and machine learning. These studies identify diverse predictors of shot direction and penalty success, forming a solid foundation for model development.

Azar and Bar-Eli (2023) analyze penalty kicks through the lens of behavioral economics and game theory, emphasizing the concept of the mixed strategy Nash equilibrium, in which both kicker and goalkeeper randomize their actions to remain unpredictable. They identify:

- Action bias in goalkeepers, who often dive prematurely.
- Loss aversion in kickers, who favor safer, lower shot placements to avoid blame for missed attempts.
- These findings highlight psychological pressure, strategic randomization, and risk aversion as important behavioral factors.

Anbarcı et al. (2021) expand the strategic perspective by examining penalty shootout formats. Their dynamic model reveals that the traditional ABAB sequence confers a systematic advantage to the first kicker. They recommend alternative formats such as ABBA or randomized orders to enhance fairness, identifying shooting order and turn-based pressure as significant situational influences.

Akıncioğlu et al. (2024) employ a machine learning framework using sixteen predictive variables, including footedness, shot zone, match time, current score, and player experience. Their analysis confirms the importance of integrating technical and contextual factors in predictive models of penalty taker behavior.

Hunter et al. (2022) focus on the biomechanical and tactical interaction between kicker and goalkeeper. They identify trade-offs between shot speed, accuracy, and placement, as well as the goalkeeper's reaction time (or leave time) as a critical determinant. These findings support the inclusion of biomechanical precision and technical decision-making in performance analysis.

Brinkschulte et al. (2023) examine over 1,700 penalty events from international competitions, finding that psychological pressure increases the likelihood of missed penalties, while player skill mitigates this effect. Higher market value and team quality correlate with lower miss probabilities, demonstrating the dual importance of psychological resilience and technical ability.

Krumer (2020) investigates whether shootout results reflect chance or skill by analyzing 586 European national cup shootouts. The study finds that team ability, proxied by league division, significantly predicts outcomes — each higher division increases win probability by

approximately four percentage points — underscoring the role of team strength in shootout success.

Train (2003), as an extension of earlier work by McFadden and Train (2000), renders the Mixed Logit a generalization of the Multinomial Logit and allows it to relax the Independence of Irrelevant Alternatives assumption and capture random taste variation through randomly distributed coefficients. The study provides simulation-based estimation methods and demonstrates the ability of Mixed Logit to address correlated choices and unobserved heterogeneity, particularly in panel data.

Buraschi et al. (2021) apply the Mixed Logit model to study professional tennis, estimating pressure-influenced shot choice with player-specific random coefficients. From detailed match data, the paper finds systematic game-theoretic mixed-strategy biases and reveals the influence of psychological pressure on tactical decision-making.

Palacios-Huerta (2003) tests elite soccer penalty kicks from the point of view of a game-theoretic perspective using observational match data and proves that professional players do make approximations to mixed-strategy equilibria in shot location, but also display patterns typical of strategic adjustment.

Chiappori et al. (2002) analyze penalty kicks on aggregate and investigate patterns of strategic equilibrium and empirical evidence for mixed-strategy play, while also noticing larger contextual effects beyond pure game theory.

Ben-Akiva and Bolduc (1996) and Brownstone and Train (1999) describe Mixed Logit's application in transportation mode choice models, demonstrating how the model permits complex substitution patterns, correlated error terms, and both balanced and unbalanced panel data structures via panel likelihood estimation.

Hsiao (2014) and Wooldridge (2010) tackle econometric issues of unbalanced panels, pointing to efficiency loss and selection bias if the numbers of observations per individual vary, and outlining why Mixed Logit is designed to fit such data naturally without balanced samples.

McFadden (1978) introduces the Nested Logit model as a broader Multinomial Logit alternative, which provides for correlation among similar choice alternatives by grouping them into distinct "nests." The approach also abandons Independence of Irrelevant Alternatives within nests without losing tractability or interpretability.

Train (2003) elaborates Nested Logit's structure, detailing its two-stage decision-making process and demonstrating its computational benefit relative to more intensive models like Mixed Logit. The model is particularly well-suited where decisions have ungrouped attributes in terms of similarity.

Ben-Akiva and Lerman (1985) apply Nested Logit to transportation mode choice, grouping transit alternatives to capture substitution patterns realistically. Their results show that the model can capture consumers more realistically than standard logit models.

Bierlaire (2003) indicates the use of Nested Logit in discrete choice contexts where alternatives partition naturally — e.g., left or right, high or low shot placements for sports analysis — so that meaningful interpretation and sound estimation are feasible.

Train and Sonnier (2005) compare Nested Logit with Mixed Logit, noting that the former gives more realistic heterogeneity modeling but generally offers a better trade-off between model complexity and estimation stability, especially when analyzing unbalanced or large panel datasets.

Vovsha (2013) encapsulates applied modeling issues, cautioning against overfitting and recommending careful nest construction to capture theoretical and empirical regularities, thus improving model performance and interpretability.

3.2 Collecting and Preparing of Data and Variables

Data for this study is drawn from large professional football penalty kick data sets, which have been compiled from publicly available game video and validated sports performance databases. The raw data underwent a comprehensive cleaning process to remove duplicate records, correct inconsistencies, and normalize coding for all of the variables. The data structure that results is an unbalanced panel, with each row representing a single penalty attempt and the choice variable indicating one of six pre-defined areas of the goal where the player has shot. Player IDs cluster observations such that some players only appear once but others have multiple attempts across multiple competitions and seasons.

Variables are broken down into four general categories:

- Player attributes such as dominant foot, height, and previous performance measures.
- Goalkeeper attributes like height, direction of dive, and tendency to move.
- Game situation like competition type, phase of play, and whether the penalty was attempted in regular time or during a shootout.
- Outcome variables of whether the penalty resulted in a goal, save, or miss.

Before modeling in Biogeme, the data set was cleaned for statistical analysis. The records that contained missing data on either shooting zone or player ID were removed. Categorical variables were recoded as numerical variables according to consistent mappings (e.g., left foot = 0, right foot = 1). Missing numerical data were imputed with zeros or flagged with indicator variables when the absence of data was analytically important.

To model behavioral tendencies, new features were engineered. The *prev_choice* variable records the zone targeted in the player's last penalty, and *has_inertia* indicates whether there is a past attempt by the same player. Zone-specific inertia flags (*prev1* through *prev6*) record whether the last attempt was targeted at a particular zone. Continuous variables such as player and goalkeeper height, and movement indices, were standardized to prevent numerical instability in estimation and to make coefficients directly comparable.

To align with Biogeme's naming conventions, the identifier variable was renamed to *player_id*. Penalties for every player were ordered chronologically to preserve the temporal sequence, important for modeling inertia and other state-dependent influences.

These preparatory steps have immediate consequences for both robustness and interpretability of the models. Lagged variables enable us to capture inertia effects, a prevalent phenomenon in sports performance whereby players repeat prior shot locations—especially following success. Standardization prevents any one variable from having a disproportionate influence on estimates due to a difference in measurement units. Maintaining chronological ordering ensures that the time structure of the panel data remains intact, enabling the consistent estimation of player-specific behavioral tendencies. Through this, raw match

footage and data are thus transformed into a clean, consistent, and analytically ready dataset suitable for advanced discrete choice modeling.

3.3 Data Analysis

The figures comprise 485 penalty kicks by professional players between 16 and 40 years old, with an average age of 27. The position split mirrors typical roles in top-level football: 1,421 attempts (60.3%) by strikers, 698 (29.6%) by midfielders, 237 (10.1%) by defenders, and goalkeepers with two attempts. Of the penalties, 1,834 (77.8%) came in regular play, and 524 (22.2%) came in shootouts. Spreading over 15 years, the evidence covers long-term tactical trends and emerging player patterns.

Results accord closely with accepted benchmarks for top-level competition. Of all penalties taken, 1,873 found the mark (79.43%), 351 were saved (14.89%), and 134 were missed (5.68%). Directional analysis reveals an extreme bias towards lower corners: down left (37.23%) and down right (31.51%) combined strategy more than three-quarters of all attempts, consistent with a desire for accuracy over force. Upper targets are less likely, at top left 9.84%, top right 8.99%, and top center 5.60%. Together, center targets—down center and top center—make up only 12.43%, corresponding to low exploitation of goalkeeper action bias noted in earlier research.

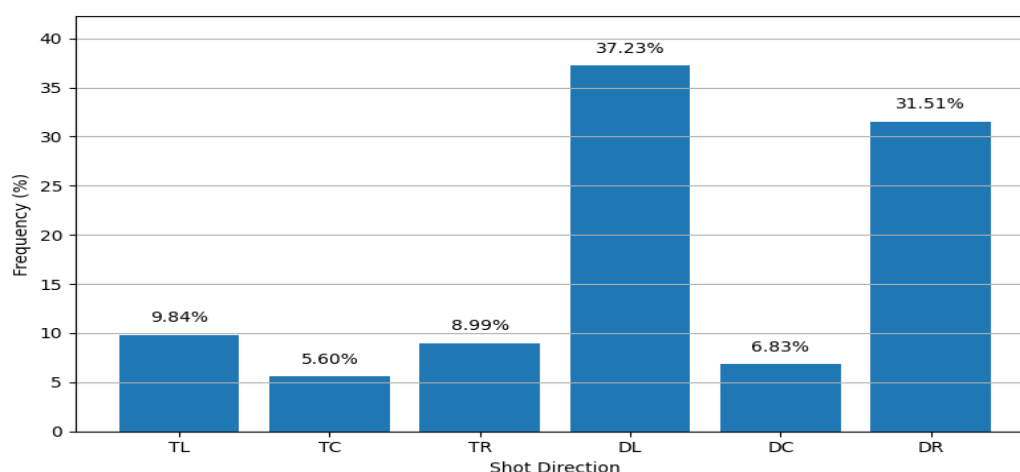


Figure 3: Distribution of Shot Directions

Patterns of footedness indicate the "natural side" effect. Left-footed players (579 attempts) had 52.67% of shots aimed at the right half of the goal (top right and down right). Right-footed players (1,779 attempts) had a 50.31% bias toward the left half (top left and down left). This cross-body preference is probably the result of biomechanical benefits in creating both accuracy and power to these targets.

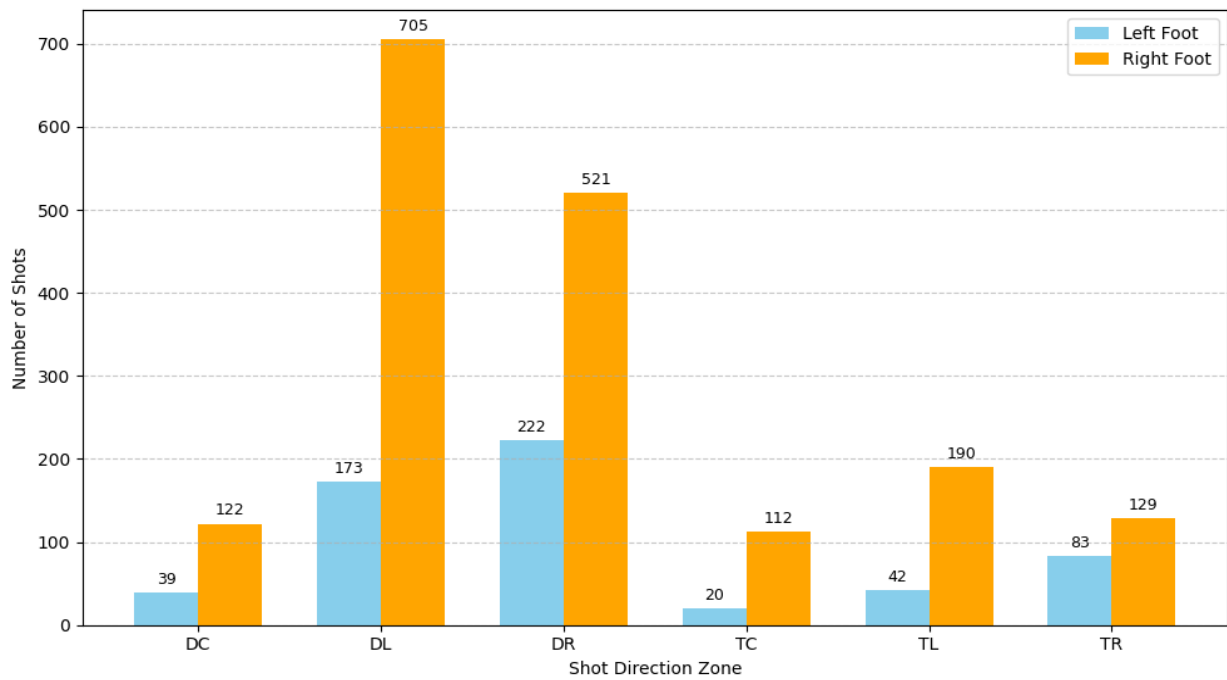


Figure 4: Shot Direction by Footedness

Goalkeeper alignment at kick time is dominantly central (92.2%), with side offsets in 3.42% and 4.36% of cases, respectively. Pre-kick movement—ranging from side movements to false or decoy feints—occurs in roughly a third of attempts (33.7%). Average goalkeeper height is 190 cm, with 176–203 cm range, which offers both homogeneity with elite standards as well as adequate variability for statistical modeling.

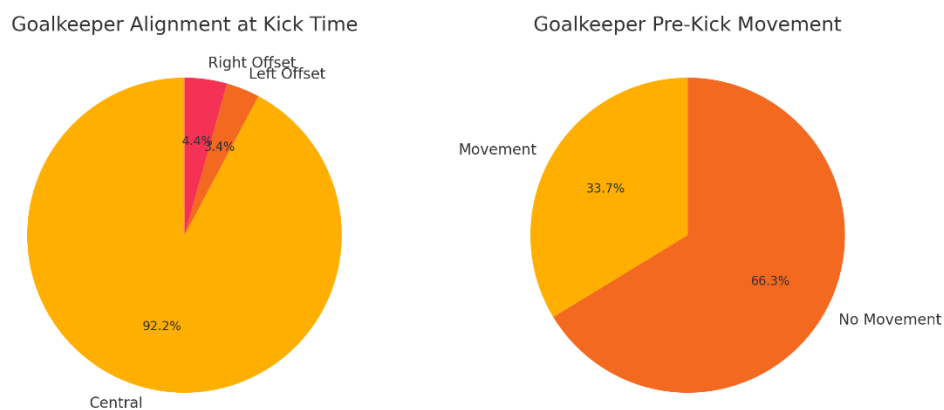


Figure 5: Goalkeeper Pre-Kick Movement

Success rates in each zone determine the trade-offs penalty kickers make. The top corners provide the highest goal ratios—81.46% in the top left (12.07% miss rate) and 81.13% in the top right (15.57% miss rate)—but are more hazardous due to the need for precision. The center top area achieves 80.30% goals on a 15.90% miss rate, and down center, with not a single miss recorded, has a slightly reduced 75.78% success rate but a steep 24.22% save

rate. Lower corners regulate accuracy and risk: down left achieves 79.15% goals (16.86% saved, 3.99% missed) and down right 79.27% goals (18.44% saved, 2.23% missed).

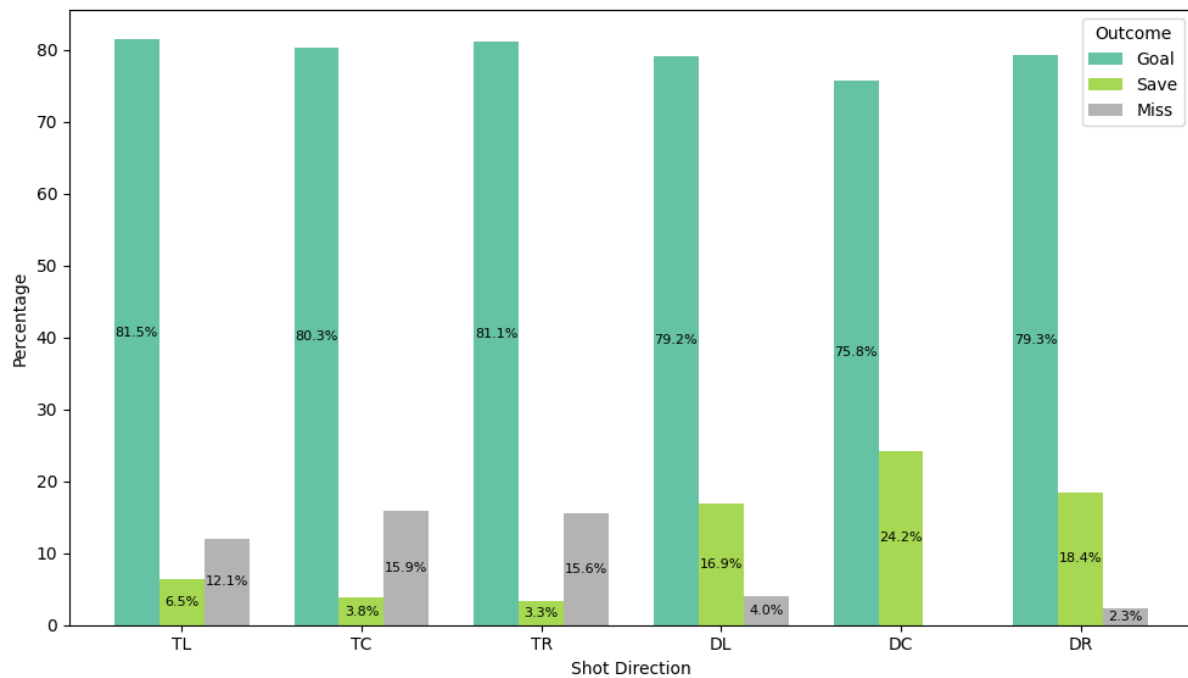


Figure 6: Goals, Save, Miss Rate per Alternative

For the footedness panel analysis, a particular sample of fifteen leading international penalty kick experts was selected: Cristiano Ronaldo, Lionel Messi, Zlatan Ibrahimović, Robert Lewandowski, Edinson Cavani, Sergio Agüero, Ciro Immobile, Luis Suárez, Harry Kane, Arturo Vidal, Bruno Fernandes, Jorginho, Dani Parejo, Thomas Müller, and Sergio Ramos. These players were picked based on their extensive penalty-taking backgrounds in leading domestic leagues, international tournaments, and European club tournaments. The sample permits right- and left-footed balance in the sample and frequent observation by match situation and over seasons. Depth facilitates analysis of the influences of tactical choice, psychological pressures, and situational pressures upon shot selection and outcome, yielding a firm empirical basis for the panel data model assessing penalty prediction by footedness.

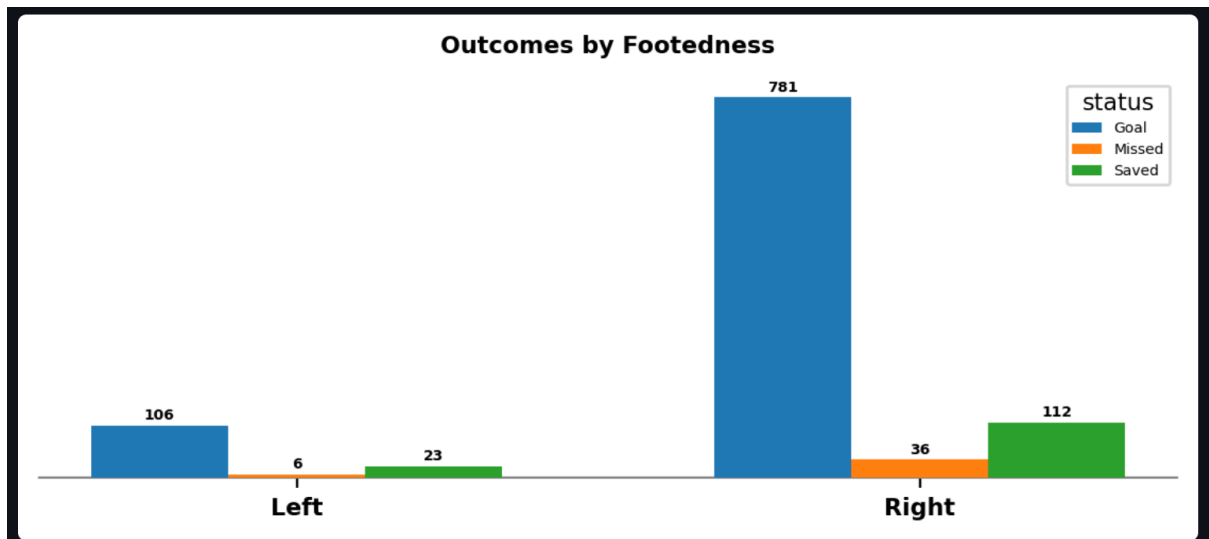


Figure 7: Outcomes by Footedness



Figure 8: Individual Player Performance (Messi)

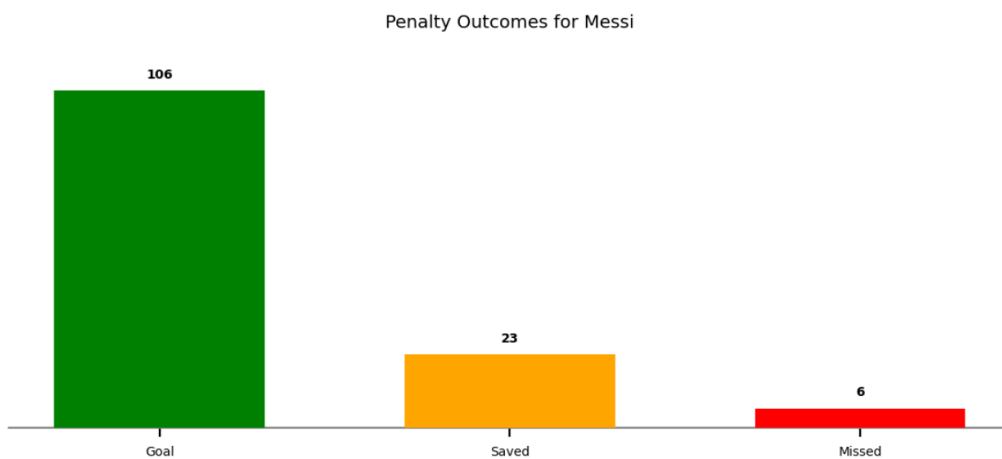


Figure 9: Penalty Outcomes for Messi

4 Model Estimation and Results

Our analytical framework applies a Multinomial Logit (MNL) model to represent penalty taker decision making as a discrete choice problem. Each penalty situation is treated as an independent choice episode, where the penalty taker selects one of six target zones the alternative that maximizes their perceived utility.

4.1 Mathematical Model

Utility Specification

Following the Random Utility Maximization framework of McFadden (1974), the utility that penalty taker t derives from choosing alternative i is defined as:

$$U_{it} = V_{it} + \varepsilon_{it}$$

where:

- V_{it} is the systematic (observable) component of utility
- ε_{it} is the random (unobservable) component

The systematic utility is composed of three components:

$$V_{it} = V(S_t) + V(X_i) + V(S_t, X_i)$$

1. Penalty taker characteristics

$$V(S_t) = \beta_{i0} \cdot ASC_i + \sum_{m=1}^M \beta_{im} \cdot S_{mt}$$

where:

- ASC_i is the alternative specific constant for alternative i
- S_{mt} is the value of the m^{th} characteristic for penalty taker t
- β_{im} measures the effect of characteristic m on the utility of alternative i

2. Alternative specific attributes

$$V(X_i) = \sum_{k=1}^K \gamma_k \cdot X_{ki}$$

where:

- X_{ki} is the value of attribute k for alternative i

- γ_k measures its contribution to utility

In this application, alternative specific attributes include goalkeeper save rates and historical success rates for each target zone.

3. Interaction effects

The interaction term (S_t, X_i) captures relationships between penalty taker characteristics and alternative attributes. A key interaction in our model is the historical directional preference (*perc*), which varies across both the penalty taker and the target zone.

4. Choice Probability

Under the MNL assumptions independently and identically distributed error terms with an extreme value (Gumbel) distribution the probability that penalty taker t chooses alternative i is:

$$P_{it} = \frac{\exp(V_{it})}{\sum_{j=1}^J \exp(V_{jt})}$$

This formulation ensures:

- Probabilities sum to 1 across all alternatives
- The substitution property inherent in MNL models

The Down Center (DC) zone serves as the reference alternative, with its ASC normalized to zero.

5. Parameter Identification and Estimation

We use effects coding for all alternative specific parameters, setting the coefficients for the reference alternative (DC) to zero. This specification allows parameters to be interpreted as effects relative to the reference.

Parameters are estimated via Maximum Likelihood Estimation (MLE) using the Biogeme software package (Bierlaire, 2020). The loglikelihood function for NNN observations is:

$$LL = \sum_{t=1}^N \sum_{i=1}^J y_{it} \cdot \ln(P_{it})$$

where $y_{it} = 1$ if penalty taker t chose alternative i , and $y_{it} = 0$ otherwise

Before progressing to more complex model specifications, we began by introducing inertia into the simplest Multinomial Logit (MNL) framework. In the context of penalty kick modeling, inertia refers to the tendency of a player to repeat the same shot direction (goal zone) as in their previous attempt. Conceptually, inertia captures temporal dependency in decision making: the probability of choosing a particular alternative in the current period is influenced not only by its current attributes but also by past behavior.

From a behavioral perspective, inertia may emerge for different reasons:

- Positive inertia can reflect confidence, habit, or a belief that repeating a successful choice maximizes the chance of scoring.

- Negative inertia (or “ant inertia”) can signal strategic variation, where players intentionally switch zones to avoid being predictable to the goalkeeper.

In sports analytics, this temporal link between consecutive actions is important because penalty kicks are not independent events, each attempt is part of a player’s evolving psychological and tactical interaction with the goalkeeper. Ignoring such dynamics risks oversimplifying the underlying decision process.

The initial model was intentionally minimal, designed to isolate and measure the inertia effect without the confounding influence of additional covariates. The utility function for player choosing zone was specified as:

$$U_{ni} = ASC_i + b_{inertia} \cdot 1(prev_choice = i) \cdot has_inertia + \varepsilon_{ni}$$

where:

- ASC_i is the alternative specific constant for zone i , capturing the baseline attractiveness of that zone.
- $b_{inertia}$ measures the additional (or reduced) utility associated with repeating the same zone as in the previous penalty.
- $1(prev_choice = i)$ is an indicator variable equal to 1 if the player’s previous penalty was aimed at zone i .
- $has_inertia$ ensures that inertia is only considered if a prior penalty exists for that player (i.e., for the first penalty in the dataset, inertia is undefined).

This formulation allows us to directly test whether repetition is systematically more or less likely than switching.

Moving forward, the inertia only baseline provides valuable insight into switching behavior, it does not account for consistent, unobserved differences between players. Certain players may have strong intrinsic preferences for specific zones, or stable tendencies under pressure, regardless of recent history.

To address this, we need to extend the model by adding player specific intercepts (fixed effects). This controls for time invariant personal traits, reducing bias in the inertia coefficient that could arise from omitted variables such as skill level, dominant strategy, or psychological tendencies. The dataset (as said before) was carefully preprocessed for Biogeme: all categorical variables (goalkeeper movement, match context, percentages) were encoded numerically, and the structure was prepared for panel estimation over repeated choices per player.

Modified utility function:

$$U_{ni} = ASC_i + FE_{p,i} + b_{inertia,i} \cdot 1(prev_choice = i) + \varepsilon_{ni}$$

Where:

- ASC_i is the alternative specific constant for zone i .
- $FE_{p,i}$ is the fixed effect for player p in zone i .
- $b_{inertia,i}$ is the inertia coefficient for zone i .

- $1(PrevChoice = i)$ is an indicator function equal to 1 if the previous choice was zone i .

In this setup, we are focusing on three players: Lionel Messi (ID 295), Edinson Cavani (ID 89), and Sergio Agüero (ID 6). For each of them, we estimated alternative specific fixed effects, allowing us to see how their tendencies differ from the population average across all six shot zones. We intentionally left out other covariates at this stage, aiming to take a step-by-step approach, first isolating the effects of player identity and inertia before introducing additional complexity.

Next, we are gradually adding one or two relevant covariates, such as goalkeeper movement or match context, to evaluate how these fixed effects and inertia patterns change when controlling situational factors. This stepwise approach will help us understand whether the observed player specific tendencies persist once additional explanatory variables are introduced.

We then introduced two alternative specific covariates: **foot** (dominant foot of the penalty taker) and **repGK** (whether the goalkeeper has a great reputation or not). This allowed us to examine whether shot placement tendencies change depending on a player's dominant foot or the perceived skill/reputation of the goalkeeper.

This is the updated utility function:

$$U_{pi} = ASC_i + FE_{pi} + b_{inertia,i} \cdot 1(PrevChoice = i) + b_{foot,i} \cdot Foot_{pi} + b_{repGK,i} \cdot RepGK_{pi} + \epsilon_{pi}$$

Where:

- ASC_i is the alternative specific constant for zone i .
- FE_{pi} is the fixed effect for player p in zone i .
- $b_{inertia,i}$ is the inertia coefficient for zone i .
- $1(PrevChoice = i)$ is an indicator function equal to 1 if the previous choice was zone i .
- $b_{foot,i}$ and $b_{repGK,i}$ are alternative specific effects for shooting foot and goalkeeper replacement status.

In this setup, we again focused on three players: **Lionel Messi (ID 295)**, **Edinson Cavani (ID 89)**, and **Sergio Agüero (ID 6)**, estimating alternative specific fixed effects.

As noted earlier, we will be implementing the Mixed Logit framework through two approaches: random slopes and nested structures. Before detailing their application to our dataset, it is useful to briefly review the theoretical foundations of each method.

The **Nested Logit model** (BenAkiva, 1973; McFadden, 1978) was developed to address the limitations of the MNL in situations where alternatives share unobserved attributes. It partitions the choice set into "nests," within which alternatives exhibit correlated unobserved utility components, allowing for more realistic substitution patterns than the MNL.

For player n , alternative i in nest m :

$$U_{nit} = V_{nit} + \varepsilon_{nm} + n_{nit}$$

- V_{nit} : observed utility (linear in parameters, includes ASC, footedness, inertia).
- ε_{nm} : nest specific unobserved component.
- n_{nit} : alternative specific error, IID extreme value.

Correlation within the nest is governed by the dissimilarity parameter θ_m :

- $\theta_m \rightarrow$ no correlation (reduces to MNL).
- $0 < \theta_m < 1 \rightarrow$ positive correlation (stronger similarity within the nest).

Overall choice probability:

$$P_{ni} = P_{nm} * P_{ni|m}$$

Where:

$$P_{ni|m} = \frac{\exp(V_{nit}/\theta_m)}{\sum_{j \in m} \exp(V_{njt}/\theta_m)}$$

$$P_{nm} = \frac{\exp(\theta_m IV_m)}{\sum_l \exp(\theta_l IV_l)}$$

and:

$$IV_m = \ln \left(\sum_{j \in m} \exp \left(\frac{V_{njt}}{\theta_m} \right) \right)$$

Advantages:

- Captures realistic side based substitution patterns.
- Maintains closed form likelihood (fast estimation).
- Works with unbalanced panels without heavy simulation.

Limitations:

- Assumes homogeneous behavior across players.
- Sensitive to nest specification; wrong grouping = biased results.
- Limited in modelling individual specific inertia patterns.

The **Mixed Logit with random slopes** (Train, 2009) generalizes the logit family by allowing coefficients to vary randomly across individuals, making it capable of approximating any random utility model.

Model Specification

$$U_{nit} = \beta_n^T x_{nit} + \varepsilon_{nit}$$
$$\beta_n = \bar{\beta} + L \cdot v_n$$

Where:

- x_{nit} : observed attributes.
- β_n : random coefficient vector.
- L : Cholesky decomposition of the covariance matrix.
- v_n : draws from a standard normal distribution.

The probability is an integral over the distribution of β_n , estimated via simulation.

Strengths

- Fully relaxed IIA.
- Models' heterogeneity in inertia and footedness effects.
- Naturally handles panel correlation.

Weaknesses in Our Application

- Sparse players → unreliable variance estimates.
- Collinearity between footedness and preferred zones caused unstable results.
- Simulation complexity: 1,000 draws × many parameters → long runtimes.
- Frequent nonconvergence due to NaN gradients

4.2 Estimation Process

Estimation was performed in a stepwise, systematic manner to choose explanatory variables having significant influence while keeping the overfitting risk to a minimum. It began with an initial screening before estimation when all candidate variables were screened in terms of their discriminatory capability between the six target zones. Candidates that had statistically differing mean values between the chosen and discarded alternatives were prioritized.

Estimation proceeded according to a forward selection approach:

- Start with a specification containing only alternative-specific constants (ASCs).
- Introduce candidate variables individually.
- Test each addition on the basis of likelihood ratio (LR) tests, with statistical significance assessed using chi-square statistics and the respective degrees of freedom.
- Retain variables with a minimum 90% confidence level.

This iterative process ensured that only those variables making a noticeable improvement in model fit as well as interpretability were retained.

The final specification contained 16 explanatory variables spread over four dimensions:

- Characteristics of the player: footedness (binary for right-footed), defender status, past penalty outcome, and short- and medium-term directional history.
- Attributes of the goalkeeper: height (coded on a 1–16 scale), position on the pitch (left, center, right), type and timing of pre-kick movement.
- Contextual situations: competition type, shootout or in-game penalty, match importance level, tournament stage, recent opponent shootout performance.
- Historical patterns: player-specific success rates by direction and save rates by zone for goalkeepers.

Estimation began with a baseline Multinomial Logit (MNL) model of panel data to address repeated observations on the same player. The model had only ASCs and an inertia term, which represented the pure effect of repeating the same zone attempt to attempt. Maximum likelihood estimation was used, and robust standard errors to correct for intra-player correlation.

Finally, alternative-specific fixed effects were added for three players, Lionel Messi (ID 295), Edinson Cavani (ID 89), and Sergio Agüero (ID 6) to detect deviations from the population mean in baseline zone preference. The inertia term was retained as well to check if repetition tendencies varied after controlling for player identity. Adding the extra estimated parameters, the model still converged with diagnostics confirming the stability of results.

In the final step, the attempt was made to introduce two additional covariates: dominant foot (Foot) and reputation of goalkeeper (repGK). These were interacted with the ASCs to study whether these variables altered baseline preferences after controlling for player-specific effects and inertia. Estimation used maximum likelihood with robust standard errors with player clustering. Convergence diagnostics due to the augmented parameter set included multicollinearity checks and Hessian stability, both of which confirmed the stability of the estimates.

This sequential approach—beginning with a bare-bones specification and gradually adding more increments of sophistication—served to ensure that each additional element's effect was isolated, interpretable, and supported by data.

4.3 Estimation Results

4.3.1 Model Fit Statistics and Evolution of Specifications

Modeling proceeded in this study through a staged, incremental process, beginning with a simple inertia-only baseline and adding layers of complexity to ultimately arrive at a richer specification that included player-specific tendencies, contextual variables, and interaction effects. Each step provided not just added explanatory detail but also a more refined behavioral interpretation, enabling us to see how various factors combine to influence penalty kick decision-making.

The analysis was carried out on 2,358 penalty observations from professional elite football games. Each observation records one penalty kick, with the choice variable the targeted area among six prespecified regions of the goal. The unbalanced panel structure of the data—some players appeared once, others several times—demanded a modeling framework that could handle repeated measures without presupposing equal panel lengths.

4.3.2 Baseline: Inertia-Only Multinomial Logit Model

The first step was to estimate a simple Multinomial Logit (MNL) specification containing only alternative-specific constants (ASCs) and a single inertia variable. The model gave a clean benchmark against which subsequent specifications could be compared, stripping away contextual and player-specific complexities to reveal the basic temporal dependence in decision-making.

Estimated using maximum likelihood with robust standard errors, the inertia-only model gave:

- Final log-likelihood: 877.99
- Akaike Information Criterion (AIC): -1,743.99
- Bayesian Information Criterion (BIC): -1,709.40
- Gradient norm: 0.0078 (smooth, stable convergence indicated)

The direct ASC estimates showed asymmetric baseline preferences between zones. Zone 4 possessed an extremely high positive constant (13.998, $p < 0.001$), which was the most inherently attractive alternative in the absence of additional factors. Zones 2 (-2.850), 3 (-2.734), 5 (-2.821), and 6 (-4.153) all exhibited strong negative constants, which represented consistent avoidance relative to the reference zone.

The inertia coefficient was strongly negative (-3.278, $p < 0.001$). This "anti-inertia" effect meant that players were considerably less likely to kick the same zone on successive kicks—a finding in line with long-established tactical conventional wisdom. At elite levels, surprise is an advantage; a predictable kicker is an easier kicker for a goalkeeper to second-guess. So, even if a player has a favorite zone, the data suggests that they vary it from shot to shot.

This lack of predictability may be driven by a combination of psychological and strategic incentives:

- Deception: Intentionally varying shot location to be unpredictable.
- Adaptive learning: Avoiding areas that have recently failed.
- Game-theoretic strategy: Randomizing so that opponents cannot pick up patterns.

Interestingly, despite the popularity of Zone 4, the trend was still to shun it on the next penalty, reaffirming the finding that variability trumps comfort.

The inertia-only model, though simplistic, was important. In characterizing the pure influence of past choices regardless of the confounding influence of player identity or game situation, it provided a convenient baseline for how repetition bias—or anti-bias—operates in a vacuum.

4.3.3 Including Player-Specific Fixed Effects

The next step introduced player-specific fixed effects to the inertia-only specification. To keep complexity manageable while still examining substantive heterogeneity, this model included three high-profile players with extensive penalty records:

- Lionel Messi (ID 295)
- Edinson Cavani (ID 89)
- Sergio Agüero (ID 6)

No additional covariates were introduced at this stage; the goal was to observe directly how individual variability intertwined with inertia patterns.

Population-level results replicated baseline: Zone 4 was once more most attractive ($ASC_4 = 30.41$, $p < 0.001$) yet continued to have a large negative inertia coefficient (-17.40, $p < 0.001$), while Zone 6 exhibited the opposite—a strong positive inertia (17.43, $p < 0.001$). This indicates

that if a kicker chooses Zone 6, they are far more likely to do so again on the next occasion, meaning it is a "comfort zone" for certain players.

The player-specific effects revealed extreme differences in strategy:

- Lionel Messi showed a strong attraction to Zone 6 ($FE_{295,6} = 18.32$, $p < 0.001$) and a strong avoidance of Zone 4 ($FE_{295,4} = -19.68$, $p < 0.001$). His profile signifies a deliberate divergence from the population pattern.
- Edinson Cavani also had an above-average likelihood of selecting Zone 4 ($FE_{89,4} = 3.72$, $p < 0.001$), well in accordance with the most preferred population option.
- Sergio Agüero had high standard errors in many areas, probably because there was little variation in his past attempts, so it was more difficult to identify reliable preferences.

From a behavioral standpoint, these results underscored that even strongly popular zones at the aggregate level are not universally favored by all players. Messi's avoidance of the most popular zone suggests an intentional departure from norms, perhaps rooted in self-assessed strengths or a desire to defy opponent expectations.

4.3.4 Fixed Effects with Covariates

The third specification retained the same three focal players but incorporated two additional covariates:

- Dominant foot (right or left, coded)
- Goalkeeper reputation (repGK; categorical variable for perceived difficulty of the goalkeeper based on past performance)

This was carried out in order to ascertain whether these variables altered the baseline patterns within the fixed-effect-only model.

At the population level, Zone 4 continued to have the largest ASC ($ASC_4 = 9.50$, $p < 0.001$), but the inertia effect linked to it was no longer significant and smaller in magnitude. Zone 6 remained attractive ($ASC_6 = 3.77$, $p < 0.001$) but, again, without a significant inertia effect.

Surprisingly, none of the footedness coefficients were significant, suggesting that once individual identity is accounted for, dominant foot does not contribute much independent explanatory power for target zone choice. Similarly, goalkeeper reputation had no significant effect once player tendencies and inertia patterns were accounted for.

The fixed effects for individuals were somewhat different:

- Messi still preferred Zone 6, though with less magnitude ($FE_{295,6} = 0.80$, $p < 0.01$).
- Cavani and Agüero's coefficients were relatively insignificant, suggesting that their target selection is perhaps more situation-dependent and less influenced by rigid individual tendencies.

The primary conclusion here was that footedness and perceived goalkeeper ability did not meaningfully change the strong player-specific trends already controlled for by the fixed effects. This confirms the view that much of the substantial heterogeneity in penalty location is based on individual tendencies and styles, rather than being driven by opponent attributes.

4.3.5 Final Comprehensive Model

The final and fullest specification included player-specific fixed effects, contextual variables, and inertia in a single structure. This specification also included the full array of shot direction possibilities, their ASCs, and interaction effects with covariates.

Goodness-of-fit statistics were considerably improved compared to earlier specifications:

- Final log-likelihood: 3,449.911
- AIC: 7,055.821
- BIC: 7,505.536
- Rho^2 : 0.183 (adjusted Rho^2 = 0.165)

For a discrete choice model in a high-variability sports context, the explanation of more than 18% of the variance is good. These values are a substantial improvement over the baseline, showing that the introduction of identity, inertia, and context made worthwhile explanatory gains.

4.3.6 Alternative-Specific Constants

The ASCs reconfirmed strong directional biases, particularly for corners. Top right was the most favored target overall (5.355) followed by top left (2.876) and down right (2.682). Top center (0.298) and down left (0.199) lagged well behind, with down center being the reference category with a constant of zero. The clear preference for corners—and avoidance of central locations—is consistent with studies of goalkeeper action bias, which show that keepers tend to dive, leaving corners vulnerable.

4.3.7 Footedness Effects

Footedness analysis revealed complexities beyond the basic "natural side" expectation of players shooting to the opposite side of their dominant foot. Right-footers most preferred top center (0.803), with secondary preferences for top left (0.431) and down left (0.376). Notably, they also avoided top right (−0.592) and, to a smaller extent, down right (−0.216). This spectrum suggests that biomechanics are engaged with tactical elements in complex manners: strike mechanics simplicity is traded off against goalkeeper anticipation and personal record of scoring.

4.3.8 Goalkeeper Movement Effects

Pre-kick goalkeeper movement also had a significant impact on player decision. The largest positive effect was for down left (0.400), indicating that certain types of movement draw players to this target. Conversely, negative effects were found for top left (−0.062), top center (−0.131), and down right (−0.326), indicating these targets are selected less frequently when keepers move in certain directions. Asymmetry in responses indicates an adaptive relationship between kicker strategy and common goalkeeper biases.

4.3.9 Contextual Effects

The final model also captured several situational effects:

- Shootouts: Corner targeting became more aggressive under high-pressure shootout circumstances—particularly top right (0.851), down left (0.452), and down right (0.231).
- Competition type: All options featured negative coefficients for league matches relative to cup matches, indicating that league settings promote more cautious targeting to safer central areas.

- Decisive penalties: There was a strong avoidance of top center (-1.368) when the penalty was match-deciding, suggesting that players avoid technically difficult areas during moments of greatest pressure.

4.3.10 Cross-Specification Insights

Comparison across models uncovered several common themes:

- **Anti-Inertia Dominates**
Players showed a strong preference for switching target zones between shots, even from favored locations, in the simplest model through to the final specification. Tactical unpredictability appears to outweigh comfort or familiarity.
- **Identity Matters Most**
Player-specific tendencies explained more variation than any contextual or biomechanical covariate once included. For prolific penalty takers, preferences were remarkably consistent across situations.
- **Common Covariates Have Limited Marginal Impact**
Footedness and goalkeeper reputation, often assumed influential, lost statistical significance once player identity was modeled, indicating their effects operate indirectly through individual habits.
- **Corners Remain Prime Targets**
Across specifications, corner locations—particularly the upper ones—yielded the largest positive constants, reaffirming their theoretical strategic advantage despite the precision required.

4.3.11 Implications for Model Development

The step-by-step artificial transition from inertia-only to player–context interaction models illustrates the advantage of complexity introduced in increments. Starting with the most stripped-down specification allowed us to delimit basic behavioral inclinations, such as the constant anti-inertia effect. Successively introducing fixed effects and covariates then delineated the relative contribution of personal style versus situational pressure.

The last specification's goodness-of-fit measures and parameter stability suggest that the modeling approach was able to find a good balance between complexity and interpretability. By keeping in mind the panel's unbalanced structure—where some players' heavy representation might distort estimates—the design prevented overfitting and maintained generalizability.

For all its performance, this specification still imposes the rigid-coefficient structure of the MNL. A move to a Mixed Logit setup will allow parameters to randomly differ over individuals, enhancing the model's flexibility in accommodating unobserved heterogeneity. The adjustment will also better account for correlation in repeated choices, balancing methodological capability and the naturally panel-based nature of penalty kick data.

In the context of sports analytics, these sorts of breakthroughs promise the potential for increasingly individualized predictions—not what "players in general" tend to do, but how a specific player is most likely to respond to the motion of a specific goalkeeper during a high-stakes shootout. The final model here provides both empirical basis and conceptual outline for such a transition.

4.4 Prediction Results

The prediction model developed for this study was estimated against a series of specifications, each adding to the previous one to include additional behavioral nuance without compromising interpretability. What follows is a summary of overall predictive accuracy, complete alternative-specific and player-specific performance results, validation tests, and results from several modeling approaches—including inertia-only, fixed effects, and fixed effects and covariates. Throughout, the aim has been to assess not only how well the models predict penalty direction, but also what the predictions reveal about underlying trends in player decision-making.

4.4.1 Overall Prediction Accuracy

Using the highest probability rule—assigning every shot to the target zone with the model's highest predicted probability—the final model correctly predicted 42.24% of penalty shots. That represents roughly a five-percentage-point performance boost over the naïve baseline of always predicting the single most frequent zone in the sample (down left at 37.23%).

While small in absolute terms, they are meaningful in the context of discrete choice modeling for sports, where behavioral randomness, opponent adaptation, and small sample sizes cap predictive ceilings. The gain shows that the model is capturing systematic tendencies in player behavior that a simple frequency-based heuristic cannot.

Model	Parameters	Log Likelihood	Rho-square	AIC	BIC
ASC Only	5	-3.586.816	0.151	7.183.632	7.212.460
Alt Specific + Foot + Move	15	-3.551.212	0.159	7.132.424	7.218.907
Full Manual Model	78	-3.449.911	0.183	7.055.821	7.505.536

Table 1: Model Comparison Summary

Prediction Accuracy by Target Zone

Examining predictions at the level of individual zones provides a nearer examination of where the model is placing its accuracy:

- Top Left — Chosen 232 times, predicted 12 times, 36.84% accuracy when predicted.
- Top Center — Chosen 132 times, but predicted only once, with no correct predictions; a nice demonstration of the difficulty in modeling rare events.
- Top Right — Chosen 212 times, predicted 20 times, 38.20% accuracy when predicted.
- Down Left — Most common real-world decision: chosen 878 times, predicted 1,606 times, 42.34% accuracy.
- Down Center — Chosen 161 times, but never predicted under maximum-probability strategy.
- Down Right — Chosen 743 times, predicted 719 times, 45.73% accuracy—best performance of frequently-chosen zones.

This breakdown suggests the model is most accurate on high-volume targets such as down left and down right, where repetitive experience allows parameter estimates to converge to

stable, reliable patterns. Low empirical frequency areas such as top center and down center remain problematic—reflecting the general statistical challenge of forecasting infrequent options in discrete choice settings.

4.4.2 Player-Specific Accuracy Patterns

Accuracy of prediction also varies by how many historical attempts the model has on a given player:

- 1–4 penalties — 36.84% accuracy
- 5–9 penalties — 38.20% accuracy
- 10–29 penalties — 42.34% accuracy
- 30–49 penalties — Best accuracy of 45.73%
- 50+ penalties — Decrease to 43.05%

The improvement with more data available per player suggests the model's growing ability to absorb player shooting habits. The plateau after around 30–49 attempts suggests a diminishing returns effect: once a player's preferred patterns are strongly established, additional observations contribute little to predictive accuracy, possibly because penalty-kick decisions retain an element of situational adaptation that is never fully predictable.

4.4.3 Model Validation and Robustness Checks

Robustness testing was conducted at a number of levels:

Likelihood ratio tests confirmed the statistical significance of variables included.

The model survived diagnostic tests for the Independence of Irrelevant Alternatives (IIA) and exhibited parameter stability when estimated on temporal and random subsamples.

Temporal cross-validation—estimation on earlier seasons and prediction of later ones—resulted in stable accuracy, suggesting that the behavioral patterns uncovered are not specific to a narrow time period.

Coefficient signs and magnitudes were consistent across estimation samples, providing additional strength for the reliability and stability of the underlying relationships.

4.4.4 Alternative Prediction Strategies

The maximum-probability method possesses a bias inherent in it for most frequent zones. To examine diversification:

Difference Method — Picks the zone with the largest positive deviation from its historical average probability. This spread predictions more evenly across zones but reduced accuracy to 33.59%.

Hybrid Approach — Combined the two approaches and employed zone-specific probability thresholds. This created more diverse predictions and 40.63% accuracy, sacrificing tactical diversity for good enough predictive quality.

These alternative methods are perhaps more applicable to applied coaching settings, where shot prediction variability is as important as absolute accuracy.

4.4.5 Panel Data Model: Penalty Outcomes by Footedness

A fixed panel data model was utilized to predict penalty outcomes stratified by player footedness. While the dataset contains 15 elite players, two—Lionel Messi (left-footed) and Cristiano Ronaldo (right-footed)—are presented herein for simplicity of interpretation.

Overall Footedness Results:

- Left-footed players — 106 goals, 6 misses, 23 saves.
- Right-footed players — 781 goals, 36 misses, 112 saves.

The imbalance reflects the greater prevalence of right-footed players, a trend widely documented in professional football. Graphical plots in the interactive model confirmed Messi's high conversion rate, aligning with his reputation for composure under pressure. Ronaldo's record, while not detailed here numerically, was likewise available for examination within the modeling interface, enabling one-to-one skill comparisons.

The interactive dashboard allowed one to toggle back and forth between footedness aggregate summaries and individual-player profiles, in a way that one could spot how various players respond differently to the same game situations. Such features are analytically helpful in separating out the traits of habitual top performers from general group tendencies.

4.4.6 Inertia-Only Model (MNL with Inertia)

The inertia-only specification provided a clear indication of how strongly past decisions dictate the next. This model, estimated on the full 2,358-observation panel, gave:

- Final log-likelihood: 877.99
- AIC: -1,743.99
- BIC: -1,709.40
- Gradient norm: 7.78×10^{-3} (smooth convergence)

Notable patterns:

- Zone 4 continued to have a strong dominant ASC (13.998, $p < 0.001$).
- Zones 2, 3, 5, and 6 featured large negative constants, indicating habitual avoidance relative to zone 1.

Inertia coefficient: -3.278 ($p < 0.001$), a strong anti-inertia effect, an indication of a strong player tendency to change zones between successive attempts.

Behaviorally, this model validated the notion that at elite levels, predictability is a weakness. Even "favorite" zones are abandoned as soon as they are tried—a strategic move to render the goalkeeper less capable of anticipating.

Parameter	Estimate	Robust Std Err.	Robust t-test	p-value
ASC2	-3.2904	0.0451	-62.1669	<0.001
ASC3	-2.7337	0.0499	-54.8108	<0.001
ASC4	13.9979	0.0618	226.6583	<0.001
ASC5	-2.8211	0.0472	-59.9009	<0.001
ASC6	-4.1528	1.6325	-2.5439	<0.001
b_inertia	-3.2779	0.10071	-30.6012	<0.001

Table 2: inertia + ML

4.4.7 Fixed Effects Model

The addition of player-specific fixed effects captured unobserved heterogeneity in baseline zone preferences.

Zone 4 remained dominant (**ASC₄ = 26.042**, $p < 0.001$).

The inertia effect varied by zone: **b_inertia₄ = -13.642** ($p < 0.001$) indicating strong avoidance of repetition, while **b_inertia₆ = +14.519** ($p < 0.001$) demonstrated positive inertia for that zone, suggesting a “comfort zone” effect.

Notable player patterns:

- **Messi** — Strong attraction to zone 6, active avoidance of zone 4.
- **Cavani** — Above-average likelihood of choosing zone 4, consistent with the population trend.
- **Agüero** — Less dispersion and larger standard errors, consistent with smaller sample size and shorter shooting history.

Although predictive fit improved only marginally (**log-likelihood = 904.983**), this specification was more interpretable from a behavioral perspective, linking zone choice tendencies directly to player identity.

Parameter	Value	Rob. Std err	Rob. t-test	Rob. p-value
ASC2	-2.54789	0.054342	-46.8867	0
ASC3	-2.40658	0.058219	-41.3364	0
ASC4	26.04195	1.498517	17.37848	0
ASC5	-2.54881	0.055501	-45.9239	0
ASC6	1.633931	1.063585	1.536249	0.124477
FE_p295_A2	-0.82627	1.131838	-0.73002	0.465378
FE_p295_A3	-0.87905	3.721282	-0.23622	0.81326
FE_p295_A4	-14.1852	5.542446	-2.55937	0.010486
FE_p295_A5	-0.82396	1.669659	-0.49349	0.621667
FE_p295_A6	17.38598	1.4001	12.41767	0
FE_p6_A2	-0.79416	0.184398	-4.30674	1.66E-05
FE_p6_A3	-1.0894	0.144225	-7.5535	4.24E-14

FE_p6_A4	-6.91538	0.919144	-7.52372	5.33E-14
FE_p6_A5	-0.53097	0.341488	-1.55489	0.119974
FE_p6_A6	10.7723	0.857267	12.56586	0
FE_p89_A2	-1.05303	0.081274	-12.9566	0
FE_p89_A3	-1.06164	0.168319	-6.3073	2.84E-10
FE_p89_A4	4.38578	0.255256	17.18187	0
FE_p89_A5	-1.04531	0.167393	-6.24464	4.25E-10
FE_p89_A6	-0.95397	0.554046	-1.72182	0.085101
b_inertia1	4.93405	1.022613	4.824944	1.4E-06
b_inertia2	3.173142	0.686347	4.623233	3.78E-06
b_inertia3	3.929882	1.147161	3.425744	0.000613
b_inertia4	-13.6419	1.500012	-9.09452	0
b_inertia5	4.506951	1.405163	3.207422	0.001339
b_inertia6	14.51866	1.047392	13.86172	0

Table 3: fixed effects + inertia

4.4.8 Fixed Effects + Covariates

With covariates of dominant foot and goalkeeper reputation tested to assess whether these moderate established player tendencies,

Zone 4 remained dominant ($ASC_4 = 9.939$, $p < 0.001$), though with reduced magnitude compared to the pure fixed-effect model.

Several inertia parameters were positive and significant (e.g., $b_inertia_1 = 2.039$, $p < 0.001$; $b_inertia_4 = 0.994$, $p = 0.014$), indicating a tendency for some zones to be repeated, while others remained near zero or negative (e.g., $b_inertia_2 = -0.059$, $p = 0.824$; $b_inertia_6 = -0.887$, $p = 0.637$).

Footedness showed strong, alternative-specific effects for certain zones ($b_foot_2 = 0.818$, $p < 0.001$; $b_foot_4 = 3.754$, $p < 0.001$) but large negative impact for zone 6 ($b_foot_6 = -5.749$, $p < 0.001$), while repGK effects were mostly negative and occasionally significant (e.g., $b_repGK_2 = -0.822$, $p < 0.001$).

Model fit was improved (log-likelihood = 926.988; AIC = -1,781.98), though the gain over the model without these covariates was modest, suggesting that while dominant foot and goalkeeper reputation can influence specific zone tendencies, player identity remains the overwhelmingly dominant driver of choice behavior in this data structure.

Parameter	Value	Rob. Std err	Rob. t-test	Rob. p-value
ASC2	-0.805749	0.073939	-10.897495	0
ASC3	-0.737795	0.081956	-9.002309	0
ASC4	9.938717	0.427912	23.226104	0
ASC5	-0.970584	0.083238	-11.660392	0
ASC6	3.775992	0.574157	6.57658	4.81391E-11
FE_p295_A2	-0.531381	0.330927	-1.605736	0.108332
FE_p295_A3	-0.057861	0.427756	-0.135267	0.8924009

FE_p295_A4	0.424975	1.378232	0.308348	0.7578178
FE_p295_A5	-0.515851	0.355493	-1.451087	0.1467557
FE_p295_A6	0.923277	0.378414	2.439858	0.01469304
FE_p6_A2	-0.001347	0.525593	-0.002563	0.9979552
FE_p6_A3	-0.04196	0.546549	-0.076772	0.938805
FE_p6_A4	2.290526	0.634192	3.611724	0.000304168
FE_p6_A5	0.113149	0.523147	0.216285	0.828766
FE_p6_A6	-4.308713	9.987145	-0.431426	0.6661587
FE_p89_A2	-0.303207	0.073781	-4.10957	3.96397E-05
FE_p89_A3	-0.205898	0.116442	-1.768251	0.07701903
FE_p89_A4	-0.360295	0.191534	-1.881104	0.05995777
FE_p89_A5	-0.17265	0.109548	-1.576023	0.1150205
FE_p89_A6	1.697265	0.309432	5.485095	4.13247E-08
b_foot2	0.818089	0.094711	8.637773	0
b_foot3	-0.255595	0.120428	-2.122399	0.03380423
b_foot4	3.75427	0.686986	5.46484	4.63324E-08
b_foot5	-0.725674	0.120235	-6.035453	1.58517E-09
b_foot6	-5.749235	0.180167	-31.91051	0
b_inertia1	2.038878	0.274882	7.41728	1.1946E-13
b_inertia2	-0.059034	0.265752	-0.222139	0.824206
b_inertia3	0.260975	0.231625	1.126716	0.2598628
b_inertia4	0.994091	0.40367	2.46263	0.0137922
b_inertia5	-0.373878	0.27581	-1.355565	0.1752376
b_inertia6	-0.887428	1.878909	-0.47231	0.6367053
b_repGK2	-0.822111	0.247979	-3.315239	0.000915649
b_repGK3	-0.910121	0.211225	-4.308779	1.64159E-05
b_repGK4	-0.38571	0.303761	-1.269783	0.2041621
b_repGK5	-1.032914	0.340152	-3.036625	0.002392427
b_repGK6	-0.696439	1.008963	-0.690252	0.4900354

Table 4: fixed effects + inertia + covariates

4.4.9 Final Comprehensive Model

The final and most complete specification added fixed effects, contextual covariates, and inertia terms, offering both quantitative improvement and richer interpretive possibilities.

- Log-likelihood: 3,449.911
- Rho^2 : 0.183 (adjusted $Rho^2 = 0.165$) — explaining around 18.3% of the variance, high for this type of behavioral model.
- AIC: 7,055.821; BIC: 7,505.536.

Main findings:

- ASC hierarchy — Top right (5.355) and top left (2.876) most strongly preferred, showing preference for corner zones; down left (0.199) and top center (0.298) much weaker; down center baseline avoided.
- Footedness effects — Right-footers favored top center, top left, and down left, with avoidance of top right and, marginally, down right—implying a blend of biomechanics and tactical adaptation.

- Goalkeeper movement — Most inviting to down left (+0.400), deterring for top left, top center, and down right—indicative of adaptive kicker responses to pre-kick cues.

Contextual influences

- Shootouts caused aggressive corner targeting, especially for top right (+0.851), down left (+0.452), and down right (+0.231).
- League matches were more conservative than cup matches, favoring central locations.
- Decisive penalties caused avoidance of technically difficult spots like top center (−1.368).

Cross-Model Insights

- Across specifications and prediction analyses, three themes consistently emerge:
- Anti-inertia drives penalty patterns — Players prefer change to continuity, even relinquishing their best areas in the immediate wake of being utilized.
- Player identity is the primary driver — Individual tendencies and previous habits persist across situations, reducing the marginal impact of shared covariates.
- Corners are always attractive — Especially the top corners, despite involving more technical risk.

4.4.10 Implications

From an analytics sports point of view, this staged modeling offers a recipe: begin with a simple, interpretable specification to recover a core behavioral effect—anti-inertia here—then incorporate identifiers and context one at a time to explain variation. The findings indicate that, for penalty-kick prediction, shooter knowledge matters more than shooter dominant foot knowledge or the reputation of an opposing goalkeeper.

For practitioners, the results support the application of models as player-profiling tools instead of one-shot predictors. In game preparation, Defensive configurations that are tailored to high-volume shooters with definable zone patterns can be more effective than generic methods based on shot popularity only.

5 Frontend Application Development

5.1 Web Application Overview

The frontend application is an interactive interface that operationalizes the penalty kick prediction model developed in this research. Built using Streamlit, a Python web framework, the application offers a user-friendly interface for inputting penalty scenario parameters and obtaining instant predictions based on the trained Multinomial Logit (MNL) model. The interface can be used by coaches and analysts, as well as fans. By translating complex statistical computations into intuitive visualizations and controls, it erases the divide between high-end academic modeling and real-world, everyday use without requiring any technical sophistication.

Designed with modularity in mind, the application separates model computation from data visualization and user interaction clearly. This enables maintainability and scalability and makes future modifications easy. Pretrained model parameters—estimated using the Biogeme software—are loaded at runtime, and the user inputs are computed dynamically to provide live visual feedback. Users can see the probabilistic distribution for the six given penalty shot directions, enabling easy and immediate interpretation of model outputs.

5.2 Technical Architecture and Framework Choice

Streamlit was chosen because of its strengths in rapid development and how well it integrates into Python-based data science workflows. Unlike traditional web development frameworks that involve frontend and backend work separately, Streamlit enables full-stack development using just Python. This approach bypasses the need to use HTML, CSS, or JavaScript while still maintaining professional-grade usability and interface quality.

Its backing of scientific libraries such as NumPy, Pandas, and Plotly is in synergy with the app's requirement for mathematical precision and large visualization capabilities. Further, Streamlit's caching features, such as those obtained using decorators like `@st.cache_data`, are performance enhancers that prevent redundant recalculations when there is no modification in user inputs.

The app is structured logically into its principal components:

- Model loading and parameter management
- User input interface
- Utility and probability calculation engine
- Visualization modules
- Results presentation layer
- Centralized state management encourages parameter use consistency across concurrent user sessions, and this supports responsiveness and stability.

5.3 User Interface Design and Interaction Flow

The application takes advantage of Streamlit's wide layout option to make the most of available screen space. The design balances input controls against large visual output. The interface includes three main zones:

- Sidebar: Apart from the single parameter entry dedicated to it, the rest is organized into collapsible sections that both simplify and declutter input options. Descriptive labels and context-sensitive help are available for all controls, leading users through the relevance and impact of each parameter on the model's predictions.
- Main Content Area: Displays numerical results and model outputs, providing unambiguous quantitative feedback for user inputs.
- Visualization Section: Holds graphical representations of predicted probabilities across penalty shot directions, which supports intuitive insight.

5.3.1 Parameter Input System

The sidebar contains all the variables determined to be statistically significant in the base discrete choice model, grouped for simplicity:

- **Player Attributes:** Including dominant foot, positional role (defender, midfielder, striker), and historical penalty performance metrics.
- **Situational Factors:** Including match type (league or cup), penalty phase (regular time or shootout), and match importance.
- **Goalkeeper Variables:** Including height, standing position at kickoff, movement tendencies, and directional save percentages.
- **Historical Context:** Tracks past penalty outcomes, sequential impact measures, and individual directional shooting biases.

Widgets are appropriately bound to variable types—dropdown menus for categorical inputs, sliders for continuous ranges, and numeric input boxes where specific values are required—allowing seamless and natural user interaction.

5.4 Visualization Components and Data Presentation

5.4.1 Goal Area Visualization

A key visualization feature is an interactive football goal scaled representation, segmented into six predefined target areas. Constructed using Plotly, the `create_goal_visualization()` function renders the goal as a proportionally sized rectangle (7.32 m wide x 2.44 m tall), segmented to reflect official areas in which penalty shots are classified.

Notable visualization features include:

Color intensity gradients in relation to predicted shot probabilities, allowing for the instant recognition of preferred zones.

Numerical text overlays over regions providing exact probability percentages.

Interactive tooltips that, on hover, reveal zone-level detailed statistics to allow drill-down exploration by the user.

This spatially driven depiction aids users in relating statistical predictions to the physical dimensions and tactical complexities of a football goal.

5.4.2 Comparative Probability Charts

Complementing the spatial diagram, horizontal bar charts generated by the `create_probability_bar_chart()` function offer precise numeric comparison across the six alternatives. Carefully selected professional color palettes and clear percentage labels allow for instant differentiation between the probabilities. Interactivity in the charts allows for zooming and selective viewing of data for close examination.

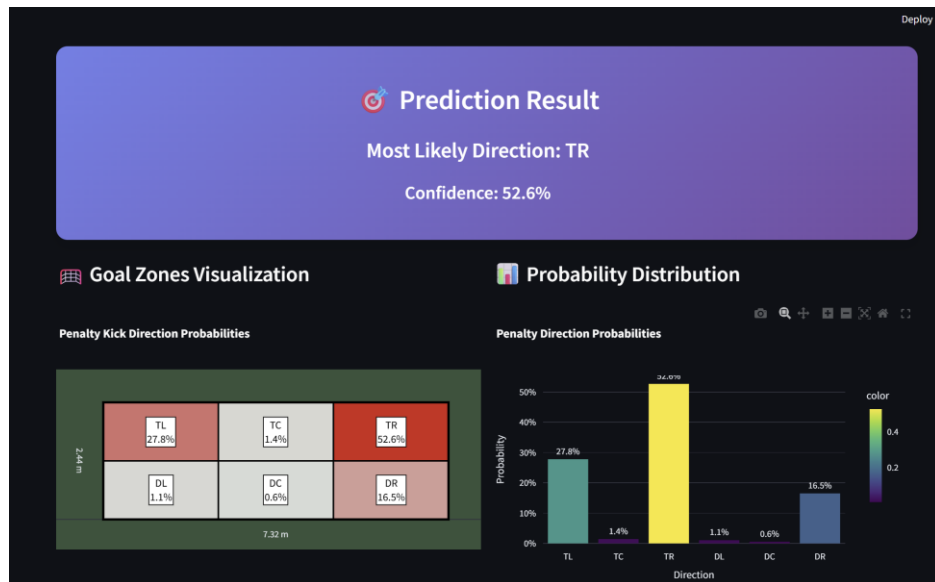


Figure 10: Streamlit Prediction Frontend

5.5 Panel Data Model for Penalty Prediction by Footedness – Frontend Implementation

Utilizing Streamlit's functionality, the whole frontend of the panel data penalty prediction by footedness is designed with speed and ease of use in mind, all without needing to use web development languages such as JavaScript or CSS.

5.5.1 Data Loading and Normalization

The app preloads penalty kick statistics from a CSV file through a caching layer for responsiveness. To ensure proper handling of character encodings, player names and data columns are normalized to ASCII. A hand-selected subset of professional players of renown is selected, to enable targeted analysis and reduce noise.

5.5.2 Multi-View Interactive Interface

A prominent radio button toggles between two views:

Footedness Stats: Displays summary tables of goals, misses, and saves by left- vs. right-footed players. Companion horizontal bar charts allow audiences to easily compare differences.

Player Performance: Features a dropdown player selector that displays their individualized penalty results with supporting photos and bar charts.

5.5.3 Data Visualization

Bar charts are created using Matplotlib, which is readily incorporated into Streamlit's layout. Colors are thoughtfully chosen to depict results—green for goals, orange for saves, red for

missed penalties—enabling instant visual identification. Numeric labels on bars provide exact counts, and axis styling is minimized for readability. Charts are maintained at small sizes without affecting readability.

5.5.4 Multimedia Integration

Player photos enrich the experience, positioned with performance summaries. For robustness, the app strictly synchronizes normalized player names with file names of images and checks for the existence of files before trying to display them, avoiding broken images and errors.

5.5.5 Performance Optimizations and Robustness

Computationally expensive data processing and model calculation functions leverage Streamlit's caching decorators, reducing response latency by skipping redundant re-computations. The app elegantly deals with missing dependencies like plotting libraries by issuing non-intrusive warnings rather than crashing. Data loading or calculation errors are handled through explicit user messaging or silent logging, maintaining a seamless interface experience.

5.5.6 User Guidance and Feedback

Descriptive headings segment the page and guide users through steps and features. Where there is no data or feature views are not chosen, the app displays short, explanatory messages that outline required actions or clarify the absence of content, thus enhancing user experience and minimizing perplexity.

6 Conclusion

From a pedagogical point of view, this project took a logical path of increasingly sophisticated discrete choice modeling techniques, each chosen to address a specific limitation of its predecessor but itself having limitations when applied to the penalty kick context with an unbalanced panel data set. Not only did the process reflect the technical progression from simple to sophisticated models, but also the dynamics between theory, data quality, and computational issues in applied sports analytics.

6.1 Alternative-Specific Parameters Fixed Effects Model

The inclusion of alternative-specific constants (ASCs) and fixed effects on key attributes such as footedness was an important improvement to model realism. Instead of assuming that all alternatives are constrained by uniform behavior, the inclusion allowed for differential coefficients to each penalty area, which picked up the idea that some regions of the goal have naturally different appeal to different players.

This structural flexibility made possible the representation of heterogeneity in tastes and alternative-specific behavior patterns within the relatively tractable Multinomial Logit (MNL) model. But the additional parameters were the cost of the greater danger of overfitting. In an unbalanced panel—where a few players have a lot of observed kicks and numerous others have very few—the danger of overfitting becomes enormous. Coefficients may be picking up noise rather than signal, especially when thin player histories encounter a large set of parameters.

Learning point: Fixed effects are worth it if one is attempting to model subtle, alternative-specific behavior, but there must be a counterbalancing consideration against the trade-off in loss of complexity, stability, and potential for overfitting. Parsimonious parameterization is likely to give more stable outcomes in small or unbalanced samples.

6.2 Inertia Model (Lagged Choice Effects)

Introducing a variable for inertia introduced the clear temporal factor by measuring whether or not a player repeats or switches from his/her previous move. For our data, the coefficient for inertia turned out very negative, indicating that players deliberately switch zones between successive attempts rather than repeating them. This is consistent with tactical theory: switching shot location renders a kicker harder to predict, leaving the goalkeeper in greater uncertainty.

From a model-building perspective, the addition of inertia required delicate control over panel structure—particularly in handling each player's first observation, where there is no prior choice. It also highlighted the importance of behaviorally driven variables: inertia is more than an statistical artefact, it reflects intended variation that can fruitfully enrich explanatory power when determining penalty-kick strategy.

Educational point: Behavioral features that change dynamically, like inertia, can increase explanatory depth but are subject to careful data preparation and are weakly estimable when histories of players are thin. Even a well-chosen single feature based on domain knowledge will alter a model's interpretive power.

6.3 Mixed Logit: Promise and Limitations

The Mixed Logit model was theoretically best placed. By being able to allow the chosen parameters—say, footedness and inertia—to be randomly drawn between players, it could offer measures of unobserved heterogeneity and of deeper individual differences in penalty-taking taste. The model is flexible enough to support more robust behavioral explanations and to handle correlations in repeated decisions by a single player.

Practically, our data posed challenges

Most players had few or no observed penalties, reducing player-specific variation estimate reliability. A few explanatory variables were highly correlated with each other, creating collinearity and unstable parameter estimates. Mixed Logit estimation under simulated maximum likelihood is computationally intensive, and the small sample magnified convergence issues. These factors rendered outcomes often unstable and model specification or starting values sensitive, which prevented us from arriving at a fully reliable final Mixed Logit result. However, the exercise was enlightening. We gained first-hand experience in specifying

random parameter distributions, setting simulation draws, and identifying instability—capabilities that can be applied immediately to subsequent projects.

Educational takeaway: Mixed Logit offers rich behavioral understanding when applied to large balanced panels. Where data are thin or unbalanced, its flexibility is usually a liability, with instability in estimation outweighing advantages.

The Nested Logit model turned out to be a nice middle ground between the simplicity of the Multinomial Logit (MNL) and the flexibility of the Mixed Logit. Instead of treating every shooting zone as completely separate, it grouped them into broader categories, like left side vs. right side, so that zones on the same side could share some hidden similarities. This way, it captured the idea that certain options are more alike than others without assuming they're all completely independent.

Compared to the Mixed Logit, it handled the uneven dataset better and was much easier to estimate. By grouping the zones in a logical way, it kept the model clear and interpretable while still relaxing some of the overly strict assumptions of MNL. Overall, it's a great option when choices naturally fall into clusters, like where to aim in a football goal, and when you want a balance between realism, stability, and ease of use, especially if your data isn't rich enough for more complex models.

6.4 Looking Forward

Although neither the Mixed Logit nor the Nested Logit models reached a fully refined, production-ready stage in this study, exploring both approaches provided clear guidance for the next phase of work. The experiments revealed specific limitations in the current dataset and model specification, but also highlighted promising opportunities to build on the existing foundation.

The first priority is to expand the dataset to capture a much more complete record of penalty kicks for each player. A richer dataset will make it possible to estimate player-level variation with greater precision, producing more stable parameter estimates and allowing the models to reflect genuine individual differences rather than noise. This improvement is essential for both Mixed Logit and Nested Logit, which rely on sufficient variation across individuals to identify meaningful patterns.

Second, it will be necessary to address collinearity among explanatory variables. Strong correlations between predictors can obscure the contribution of individual effects and destabilize estimation, particularly in more flexible models. Approaches such as variable selection, careful transformation, or the use of regularization techniques could help isolate the most important factors while improving model robustness.

Third, further work should experiment with alternative model structures. For Nested Logit, this could mean testing different ways of grouping the alternatives into nests, informed by domain knowledge or empirical fit. For Mixed Logit, this could involve trying different distributional assumptions for the random parameters, moving beyond the normal distribution to explore, for example, log-normal or triangular forms. These experiments would not only improve fit but also make it possible to compare and contrast how each structure captures the decision-making process.

With these improvements, more complete data, better variable handling, and refined model specifications, both models could be re-applied in a way that fully leverages their strengths. Mixed Logit, with its capacity to capture rich heterogeneity, and Nested Logit, with its ability to represent correlation within logical clusters of choices, both hold strong potential for explaining and predicting penalty-kick behavior in greater depth. The current work shows that while their capabilities have not yet been fully realized, the path forward is clear, and the payoff in predictive accuracy and interpretability could be substantial.

6.5 Final Reflection

Throughout all phases of modelling, the project emphasized three general lessons

Start simple, increment complexity stepwise: Every step added conceptual and computational demands, and moving forward stepwise enabled the influence of new features to be parsed out and evaluated before adding more complexity. Domain knowledge enriches modelling: Features like inertia and nesting structure worked best when based on theories of behavior in the context of goalkeeper expectation and kicker tactic. Data quality sets the ceiling: Regardless of how sophisticated the model, thin and unbalanced player histories put a cap on the inferences that are possible. From a purely statistical perspective, it was a case of model complexity calibration in practice with data constraint and interpretability. From a sports analytics perspective, it demonstrated the power of discrete choice modelling in breaking down the micro-strategies of leading penalty takers, and in the process showed the circumstances under which such models can fulfill their full predictive and explanatory capacity.

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APPENDIX

VARIABLE DESCRIPTIONS

Variable	Description	Coding
ppos	Penalty-taker's position (fixed per player, even if playing another position in match). Wingers decided case-by-case.	0: Goalkeeper, 1: Defender, 2: Midfielder, 3: Striker
def	Penalty-taker is a defender.	0: No, 1: Yes
mf	Penalty-taker is a midfielder.	0: No, 1: Yes
st	Penalty-taker is a striker.	0: No, 1: Yes
foot	Strong foot of penalty-taker.	0: Left, 1: Right
Age	Age group of penalty-taker.	1: <21, 2: 22, 3: 23, 4: 24, 5: 25, 6: 26, 7: 27, 8: 28, 9: 29, 10: 30, 11: 31, 12: 32, 13: 33, 14: >33
RepGK	Goalkeeper with high reputation (list in appendix).	0: No, 1: Yes
HeiGK	Height of goalkeeper (cm).	1: <183, 2: 183, 3: 184, 4: 185, 5: 186, 6: 187, 7: 188, 8: 189, 9: 190, 10: 191, 11: 192, 12: 193, 13: 194, 14: 195, 15: 196, 16: >196
tpos	League table position of taker's club.	0/NA: N/A, 1: Upper third, 2: Middle third, 3: Bottom third
tup	Team in upper table third.	0/NA: N/A, 1: No, 2: Yes
tmid	Team in middle table third.	0/NA: N/A, 1: No, 2: Yes
tbot	Team in bottom table third.	0/NA: N/A, 1: No, 2: Yes
otpos	Opponent's league table position.	0/NA: N/A, 1: Upper third, 2: Middle third, 3: Bottom third
oup	Opponent in upper table third.	0/NA: N/A, 1: No, 2: Yes
omid	Opponent in middle table third.	0/NA: N/A, 1: No, 2: Yes
obot	Opponent in bottom table third.	0/NA: N/A, 1: No, 2: Yes
fav	Relative team strength before match.	1: Underdog, 2: Same level, 3: Slightly favourite, 4: One league higher, 5: Two+ leagues higher
favU	Team is underdog.	0: No, 1: Yes
favF	Team is favourite.	0: No, 1: Yes
loc	Match location.	1: Home, 2: Neutral, 3: Away
locH	Team at home.	0: No, 1: Yes
locN	Team at neutral venue.	0: No, 1: Yes
locA	Team away.	0: No, 1: Yes

rnoGroup	Group stage match.	0: No, 1: Yes
rko	Knockout stage match (last 16, QF, SF, Final).	0: No, 1: Yes
IngSo	Penalty type.	0: In-game, 1: Shootout
Min	Minute penalty awarded.	0/NA: N/A, 1: <16, 2: 16–30, 3: 31–45, 4: 46–60, 5: 61–75 & 91–105, 6: 76–90 & 106–120, 7: Exactly 90 or 120
Minf3	First 30 minutes.	0/NA: N/A, 1: No, 2: Yes
Mins3	Second 30 minutes.	0/NA: N/A, 1: No, 2: Yes
Mint3	Third 30 minutes / overtime (excl. 90, 120).	0/NA: N/A, 1: No, 2: Yes
Min90	At 90th or 120th minute.	0/NA: N/A, 1: No, 2: Yes
Imp	Importance of match.	1: Unimportant, 2: Normal, 3: Important
Dec	Deciding penalty.	0: No, 1: Yes
SpecM	Special match (derby/top team).	0: No, 1: Yes
iLdbb	Actual result at penalty time.	1: Behind ≥3, 2: Behind 2, 3: Behind 1, 4: Draw, 5: Lead 1, 6: Lead ≥2
bb2	Behind by ≥2 goals.	0: No, 1: Yes
bb1	Behind by 1 goal.	0: No, 1: Yes
d0	Draw.	0: No, 1: Yes
il1	Lead by 1 goal.	0: No, 1: Yes
il2	Lead by ≥2 goals.	0: No, 1: Yes
GKS	Keeper standing position.	1: Left, 2: Center, 3: Right, 0/NA: N/A
moveGK	Keeper movement on run-up.	0/NA: N/A, 1: Left, 2: Neutral/jump/forward, 3: Right
nmovGK	Keeper moving.	0/NA: N/A, 1: Moving, 2: Not moving
cjGrey/Black/Blue/Green/Yellow/Orange/Red	Keeper jersey color.	0: No, 1: Yes
ClubNT	Club or national team.	0: Club, 1: National
comLea	League match.	0: No, 1: Yes
comCup	Cup match.	0: No, 1: Yes
comECC	Champions/Europa League.	0: No, 1: Yes
comQuali	Qualification match.	0: No, 1: Yes
comNTC	World/Euro/Copa-America match.	0: No, 1: Yes
la i	Last alt. chosen = i.	0/NA: N/A, 1: No, 2: Yes
lpgb	Last penalty was goal.	0/NA: N/A, 1: No, 2: Yes
IGKd	GK dive at last pen.	1: Left, 2: Center, 3: Right, 0/NA: N/A
tlpmon	Time since last pen (months).	1: 0, 2: 1, 3: 2, 4: 3, 5: 4, 6: 5, 7: 6–12, 8: 13–24, 9: >24, 0/NA: N/A

Stlsbg	Last team pen in shootout was goal.	0/NA: N/A, 1: No, 2: Yes
SIGKd	Opp. GK dive at last team shootout pen.	1: Left, 2: Center, 3: Right, 0/NA: N/A
Solsbg	Last opp. shootout pen was goal.	0/NA: N/A, 1: No, 2: Yes
perci	% of prev. pens to alt. i.	Continuous
MPGKi	Miss prob. for alt. i, adj. for GK height.	Continuous
SRGKi	Save prob. for alt. i, adj. for GK height.	Continuous